

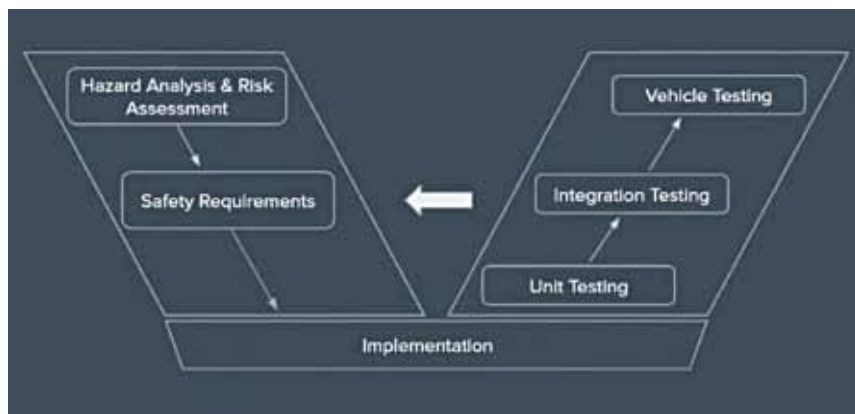


Fig. 3: V-model of safety design and requirements

be made, testing the overall system, and simulating the function in all scenarios.

Another goal is to minimise area 3 (unknown, unintended scenarios) to an acceptable level. Contributing measures for minimising it include endurance testing, driving test, close-course testing, and virtual simulation.

Cybersecurity. Extensive connectivity among automated driving vehicles, infrastructure, and pedestrians poses a challenge to protect fleets and customers from cyberattacks. The shift in vehicle autonomy from L2 to L3/L4 has made the automated driving functionalities critically reliant on external data that consists of sensor data, maps, localisation and positioning information. If the data integrity gets compromised, the automated driving function will use faulty data to manoeuvre the vehicle, resulting in inaccurate driving. With respect to cybersecurity, safety and security are tied in together. And it needs to be highly-robust.

Verification and validation of implemented design

In Fig. 3, the left-hand side is concerned with the design and requirement specifications. Verification of the safety requirements ensures that the known scenarios get covered. It may also lead to functional design improvements. Verification is an iterative process, which increases confidence in safety.

On the right side in Fig. 3 are different testing levels consisting of unit testing, integration testing (where different components are integrated), and vehicle level testing.

While the principle of safety by design is fundamental, it remains insufficient due to the inability to foresee unknown, unintended scenarios. Therefore, validation aims at confirming safety across both known and unknown scenarios with enough confidence.

Validation tests a verified automated system in uncertain scenarios that it would likely encounter with the help of closed-course testing or virtual environment simulation. Similar to verification, validation may also trigger changes to the functional design. It is also an iterative process and eventually increases confidence in safety.

Gearing up for the unanticipated

Thanks to the enhancements in electronics, especially sensors and computational systems, vehicle safety has witnessed a huge stride over the past few years. Although this has brought a great deal of comfort for people who drive, maintaining it in abrupt situations is quite challenging.

The good news is it can be reduced to a great extent by adhering to key safety principles that prove decisive in such conditions. Smart automotive safety can only be met when design and intelligence work hand-in-hand. **EFY**

The article is based on the talk 'Autonomous Vehicle Safety Overview - Challenges and Key Principles' by Palak Talwar, Senior Safety Engineer, Lyft Level 5, which was presented at June edition of Tech World Congress 2021. It is prepared by Vinay Prabhakar Minj, a technology journalist at EFY

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